BUSINESS CASE

Car Insurance: Risk Segmentation (A Statistically Confirmed Null Result)

1. Executive Summary

None of the six policyholder or vehicle attributes tested coverage zone, marital status, education, car colour, gender, and car use show a statistically significant relationship with claim amount. Every one-way ANOVA test returns $p > 0.05$, and correlations between household income or vehicle year and claims are effectively zero ($r < 0.005$ in every case). This is a confirmed null result, backed by formal statistical testing, not an eyeballed impression.

Recommendation: the business should not price or underwrite based on any of these six variables. The honest next step is investing in real risk data driving record, annual mileage, prior at-fault accidents, credit-based insurance score none of which exist in the current dataset.

2. Business Problem

The business wants to know which customer or vehicle attributes predict insurance claims, in order to inform pricing and underwriting decisions.

3. Data & Method

37,542 policies were tested across six variables. For categorical variables (coverage zone, marital status, education, car colour, gender, car use), a one-way ANOVA was run against claim amount. For continuous variables (household income, vehicle year), Pearson correlation was calculated against both claim amount and claim frequency. A minimum sample size of 30 policies was enforced before reporting any car make-level average, after an initial pass showed the largest apparent differences sitting on samples of 2–10 policies noise, not signal.

4. Key Findings

Variable	Test	Result	Verdict
Coverage zone	ANOVA	$F = 0.51$, $p = 0.728$	Not significant
Marital status	ANOVA	$F = 1.33$, $p = 0.263$	Not significant
Education	ANOVA	$F = 1.47$, $p = 0.220$	Not significant
Car color	ANOVA	$F = 0.86$, $p = 0.625$	Not significant
Gender	ANOVA	$F = 1.30$, $p = 0.255$	Not significant
Car use	ANOVA	$F = 0.01$, $p = 0.938$	Not significant
Household income	Pearson correlation	$r = -0.005$, $p = 0.364$	No relationship
Vehicle year	Pearson correlation	$r = +0.001$, $p = 0.816$	No relationship

A secondary check on car make initially showed one make averaging ₦79,049 in claims but that make had only 3 policies. Applying a 30-policy minimum sample size, the range across all trustworthy makes narrows to ₦45,419–₦56,268, still consistent with no real effect once small samples are excluded.

5. Root Cause

The dataset captures demographic and vehicle-cosmetic attributes rather than actuarial risk indicators. Claims in the real world are driven by driving behaviour and exposure (mileage, prior accidents, credit-based insurance score) none of which this dataset records. The absence of signal is therefore explained by what the data does and doesn't measure, not by an error in the analysis.

6. Recommendation

Do not build a pricing or underwriting model on the six variables tested here formal testing confirms none of them separate high-risk from low-risk policyholders.

Request that data engineering add real risk variables (driving history, annual mileage, prior at-fault claims, credit-based insurance score) before attempting any segmentation or pricing model.

In the meantime, keep pricing flat across these attributes rather than justifying rate differences the data does not support doing otherwise risks regulatory and fairness exposure with no offsetting accuracy benefit.

7. How to Validate This Before Acting

If new variables are added to the dataset, re-run the same ANOVA/correlation battery before assuming they'll perform better test, don't assume.
Any future car make or car model analysis should keep the 30-policy minimum sample size rule to avoid re-introducing small-sample noise as a false finding.

If a pricing model is eventually built on new risk variables, this null result serves as the baseline it needs to beat.